ENTRY DYNAMICS

Session 17

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Entry Dynamics

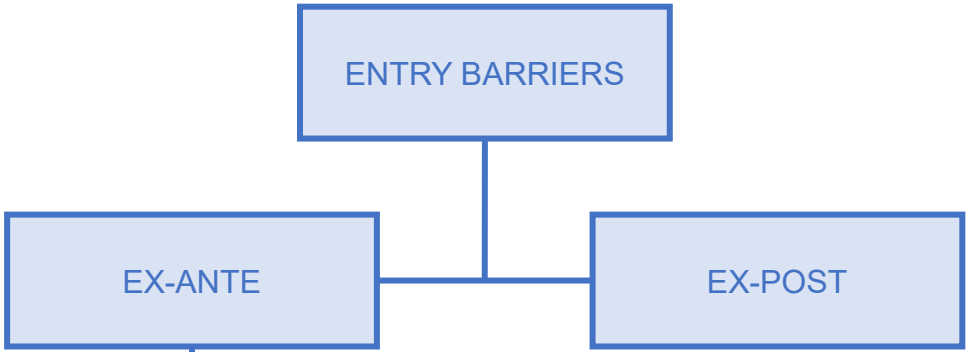
Barriers to Entry (2m.). https://www.youtube.com/watch?v=Jx8fEumsbQo



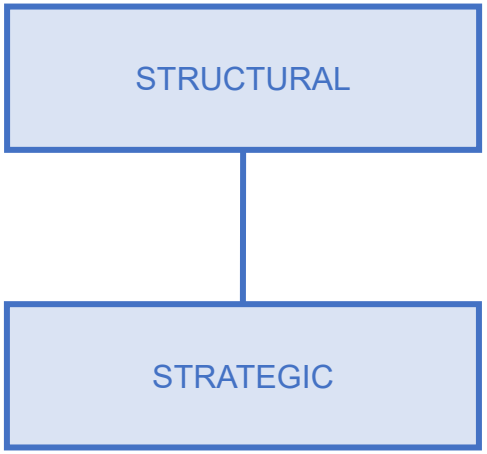
1. **Barriers to Entry**
2. **The three Bain´s entry conditions**
   - Blockade Entry
   - Accommodating Entry
   - Deterred Entry
3. **Typology of Entry Conditions – Fundenberg & Tirole matrix**
4. **First Mover Advantage**
5. **Asymmetry of Information and Signaling Theory**

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Ex-ante vs. Ex-post



Structural vs. strategic



Barriers to **entry** are the **obstacles** that make it difficult for new companies to enter a given industry, segment or market.

Structural barriers → Barriers that can`t be prevented but occur naturally. Depends on the industry or segment.

Ex.: Regulation - Pharma Industry

Strategic barriers → Behavioral, consist of a decision to be aggressive or accommodating.

Ex.: Pricing Generics drugs – Pharma industry

Entry Barriers

Ex-ante:

Structural :

Exogenous (Nature of the industry)

- Government regulations
- High Capital Requirements
- Economies of scale & scope
- Resources → Control of essential resources by incumbent
- Low Demand segments or industries

Endogenous (Raised by incumbents)

- Technological lead (Investments in R&D, patents)
- Marketing investments (Customer loyalty)

Strategic:

- Limit Pricing
- Strategic Bundling
- Pre-emption

Ex-Post:

Strategic:

- Predatory pricing
- Capacity Expansion

Entry Dynamics

Strategic barriers to entry. (2m.) https://www.youtube.com/watch?v=MVmj7qe1_aU



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Entry Barriers + Ex-ante + Structural + Exogenous (Nature of the industry)

- Government regulations

Ex.: TELCO, Electricity, Banks, Investment agencies… needs government authorization

- High Capital Requirements

Ex.: Automobile industry, Airplanes industry, Bank industry, ….

- Economies of scale & scope

Economies of scope → Reduce the cost per unit by increasing the production of different products or services.

Economies of scale → Reduction of the unit price by increasing the production of the same product /service.

Ex.: Diversification Strategy (Economies of Scope); Steal industry (Economies of scale)

- Resources → Control of essential resources by incumbent

Ex.: Diamonds industry, Top software engineers for Google or Apple, etc.

- Low Demand segments or industries

Ex.: Specific cancer drugs or Rare disease (A lot of I+D and not lot of customers)



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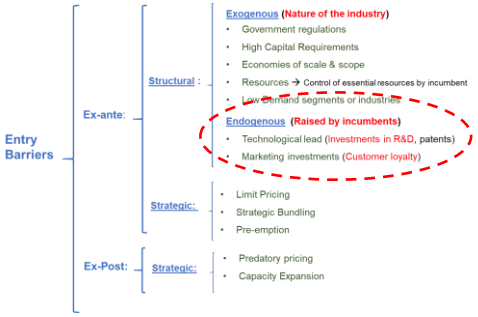
- Predatory pricing
- Capacity Expansion

Entry Barriers + ex-ante + Structural

+ Endogenous (Raised by incumbents)

- Technological lead
  - R+D+I; Patents; New process design.

Ex.: PHARMA industry; TELCO industry; Automobile industry



- Marketing investments
  - Branding; customer centricity; Luxury….
  - Increase the customer loyalty.

Ex.: Hermes; Loewe; Ferrari; Apple …

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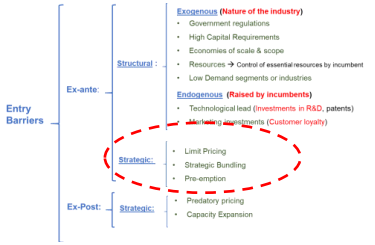
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Entry Barriers + ex-ante + Strategic



- Limit Pricing
  - is the practice of setting a product price at a level just low enough to deter potential market entrants from competing in a market.

Ex.: WhatsApp (0 €), Instagram (0 €), Low cost airlines; etc…

- Strategic Bundling
  - is strategy where companies sell several products or services together as a single combined unit.

Ex.: TELCOS (Tel + Mobile + TV); Electric companies (GAS + Electricity)

- Pre-emption
  - First option to buy before it can be offered to any other person or entity.

Ex.: Microsoft software partners; Oil companies with their own oil reserves; etc…

Cities / Countries offer this possibility to local strategic industry or financial industry.

Entry Barriers

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Strategic:

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Ex-Post:

Strategic:

- Predatory pricing
- Capacity Expansion

Entry Barriers + Ex-post + Strategic



- Predatory pricing
  - is a pricing strategy in which goods or services are offered at a very low price point, with the intention of driving out competition and creating barriers to entry.

Ex.: international dumping exports; Airlines price competition; etc…

- Capacity Expansion
  - Growth + Achieve as much market share as possible

Ex.: Amazon, Google, Microsoft, Netflix → Capture as much market share as possible

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What is the difference between "Limit pricing" and "Predatory pricing"?

1. Barriers to Entry

2. **The three Bain´s entry conditions**
   - Blockade Entry
   - Accommodating Entry
   - Deterred Entry

3. Typology of Entry Conditions – Fundenberg & Tirole matrix

4. First Mover Advantage

5. Asymmetry of Information and Signaling Theory

The three Bain entry conditions



When potential rivals wants to enter there are three different possibilities:

1. Blockaded entry

Existing structural barriers are effective in deterring entry.

2. Accommodated entry

Structural barriers may be low and strategic barriers are ineffective deterring entry or they are not cost effective.

3. Deterred entry

Entry deterring strategies are effective in discouraging potential rivals and are cost effective.

1. Blockaded entry

(Spanish:"entrada bloqueada")

1. Blockade Entry
2. Accommodating Entry
3. Deterred Entry

Existing structural barriers are effective in deterring entry

High costs at entry, undifferentiated product for which it is difficult to rise price above marginal costs…

REACTION → Incumbent does not need to take any action to deter entry (superfluous effort)

Ex. Car Industry; Steel Industry, etc…

1. Blockade Entry
2. Accommodating Entry
3. Deterred Entry

2. Accommodated entry

(Spanish:"entrada comoda")

Incumbents should not bother to deter entry

E.x.: Typical in markets with growing demand or rapid technological change…

WHY? →

1. Structural barriers are low.
2. Strategic barriers are ineffective.
3. The incumbent's costs to deter entry exceed the benefits (wasted effort)

   Ex.: Software games; ecommerce.

3. Deterred entry

(Spanish:"Disuadir la entrada")

1. Blockade Entry
2. Accommodating Entry
3. Deterred Entry

- Entry is not blockaded. It is attractive to enter these markets.
- Entry deterring strategies are effective in discouraging potential rivals and pay off for rivals (cost effective)
- Deterred entry is the only condition under which incumbents should engage in deterring acts.
  - Strategies to deter entry:

1) Strategic Bundling

2) Predatory pricing

3) Limit pricing

Bundling is strategy where companies sell several products or services together as a single combined unit.
Ex.: TELCOS → Tel + Movil + Wire TV

Predatory pricing: is a pricing strategy in which goods or services are offered at a very low price point, with the intention of driving out competition and creating barriers to entry.
Ex.: Dumping exports

Limit pricing is the practice of setting a product price at a level just low enough to deter potential market entrants from competing in a market.
Ex.: WhatsApp, Telegram, Instagram, etc…

1. STRATEGIC BUNDLING

Deterred Entry

- A combination of goods or services is sold at a price that is less than what it would cost to buy the same items separately.

Ex.: Amazon; Vacation packages, NetFlix bundles, DVD and internet streaming.

2. LIMIT PRICING → Discourage NEW competitors

- Incumbent charges a low price to discourage new firms from entering
- If an incumbent sets the limit price low enough, a potential entrant will conclude that post-entry profits will not cover the sunk costs of entry and therefore will not enter

3. PREDATORY PRICING → Against CURRENT competitors

- Incumbent sets a low price to drive smaller rivals from the market (by making it unprofitable for rivals to stay)
- The purpose is two: (1) Drive out current rivals. & (2) Deter future rivals from entering



VS



Example

Prescription drug price war

- Walmart, seeking to undercut the competition, initially began offering certain prescription drugs at well below its price floor.
- Target, looking not to be undercut, matched these drug price cuts.
- However, Minnesota state law forbids the sale of drugs below their stated cost and limited the discount, thus putting an end to the **price war**.

→ Price floor: Set by the government, a price floor is the *lowest price that goods or commodities can be legally sold* at based on the minimum cost at which turning a profit is still possible.

✓ Deterring Strategies are under regulation. See here for instance, a report by OECD
✓ https://www.oecd.org/competition/abuse/2375661.pdf

Is being Aggressive paying-off?

Analyzing **game theory suggest: → NO;** *Remember the Lobster Game*

- Why? Because the incumbent's pre-entry pricing does not influence the entrant's expectations about post-entry competition.

But → Why may limit pricing still work? → IT MAY WORK

In real life knowledge about industry conditions if often asymmetric. For example:

1. Entrants don´t know IF THE DEMAND IS MEETING THE EXPECTATIONS of customers.

   Entrant might not be certain whether the incumbent limited the price and whether the market demand is "meeting expectations" or is "below expectations"

2. Entrants don´t know THE COSTS → incumbents knows the costs

   Entrant might be uncertain whether the incumbent has "typical costs" or "low costs"

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Typology of Entry Conditions

(Fudenberg & Tirole, 1984)

Profitability of entry depends on ex-ante and ex-post strategies of incumbents

		Ex-post Entry Barriers	
		Aggressive (Retaliation) (Spanish: "Represalias")	**Accommodating**
Ex-ante Entry Barriers	**High**	Top Dog	Fat Cat
	Low	Lean & Hungry	Puppy Dog



TOP DOG

		Ex-post Entry Barriers	
		Aggressive (Retaliation) (Spanish: "Represalias")	Accommodating
Ex-ante Entry Barriers	High	Top Dog	Fat Cat
	Low	Lean & Hungry	Puppy Dog

- **Top Dog**: Difficult for new entrants, usually dominated by big firms

  E.g.: Pharma if there is possibility of competition - high advertising expenses,

  But they don´t compete in generic pharma (low cost pharma).

  E.g.: Oligopoly industries in general

- Economies of Scale
- Price Wars







FAT CAT

		Ex-post Entry Barriers	
		Aggressive (Retaliation) (Spanish: "Represalias")	Accommodating
Ex-ante Entry Barriers	High	Top Dog	Fat Cat
	Low	Lean & Hungry	Puppy Dog

- **Fat Cat**: Entry tough but less competitive than Top Dog

  E.g.: Construction, Luxury Goods, Ship Building

  - Economies of Scale
  - No price wars



ARMANI



LEAN & HUNGRY

		Ex-post Entry Barriers	
		Aggressive (Retaliation) (Spanish: "Represalias")	Accommodating
Ex-ante Entry Barriers	High	Top Dog	Fat Cat
	Low	Lean & Hungry	Puppy Dog

- **Lean & Hungry**: Entry is easy, but sector is highly competitive and aggressive.

E.g.: Magazines - reader loyalty,

Restaurants (not luxury), fast food,…)

- Not high investments (nor in R&D r
- Very likely to have price wars





PUPPY DOG

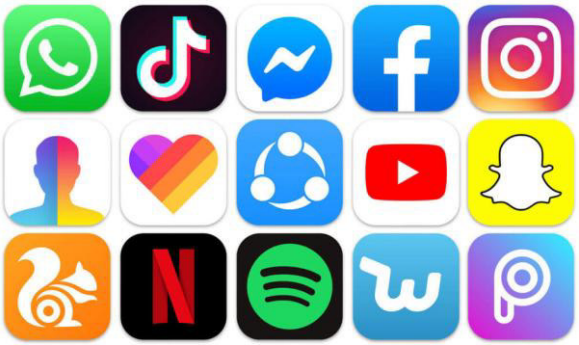
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- **Puppy Dog**: Entry easy and competition is low

→ Some new Tech industries; Law firms; middle consultancy firms

E.g.: software / app – service

- Not high investments required
- Scarce competition







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First mover advantage

- In the game: "moving first" generates an advantage, except if both players have a dominant strategy.
- More generally (beyond games) First mover ADVANTAGE are typically:
  - Learning Curve
  - Reputation and Buyer Uncertainty
  - Switching Costs



- But being the first mover can be a DISADVANTAGE as well:
  - Lack of understanding of: technology, customers, market, etc…
  - Failure of the initial technology
  - Lack of complementary assets

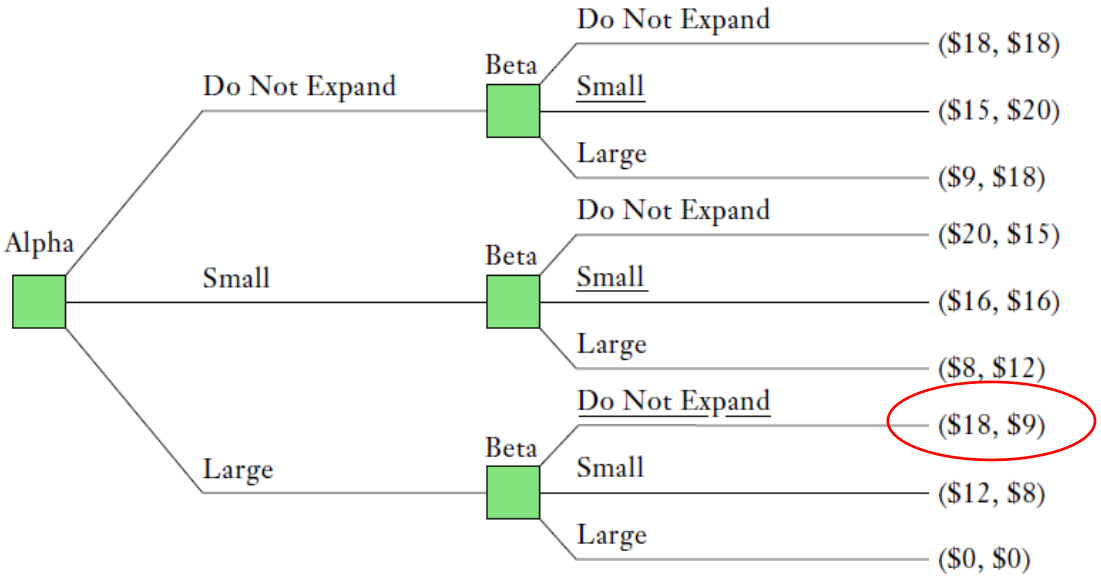
When it is interesting to be the FMA?

First mover advantages are clustered into 3 main ones: (*Lieberman & Montgomery, 1988*):

1. **Technology Leadership**: exploitation of learning curve
2. Ex.: Mobil phone industry; Pharma industry patents; etc.
3. **Control of Resources**: ability to control strategic and scares resources
4. Ex.: Oil industry; Electric car industry; Airport slots for airline industry; etc.
5. **Buyer Switching Costs**

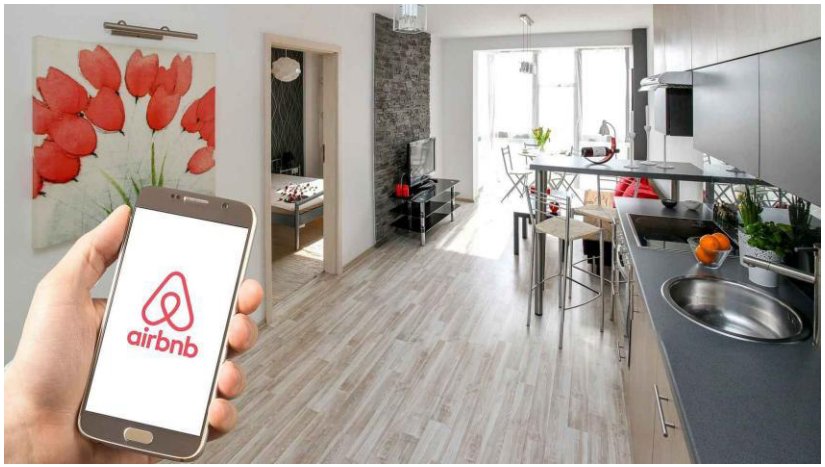
   Ex.: WhatsApp; Apple; PDF Adobe; CANON vs. NIKON; etc.

		Beta		
		Do Not Expand	*Small*	*Expand*
Alpha	DO NOT EXPAND	\$18, \$18	\$15, \$20	\$9, \$18
	SMALL	\$20, \$15	\$16, \$16	\$8, \$12
	LARGE	\$18, \$9	\$12, \$8	\$0, \$0



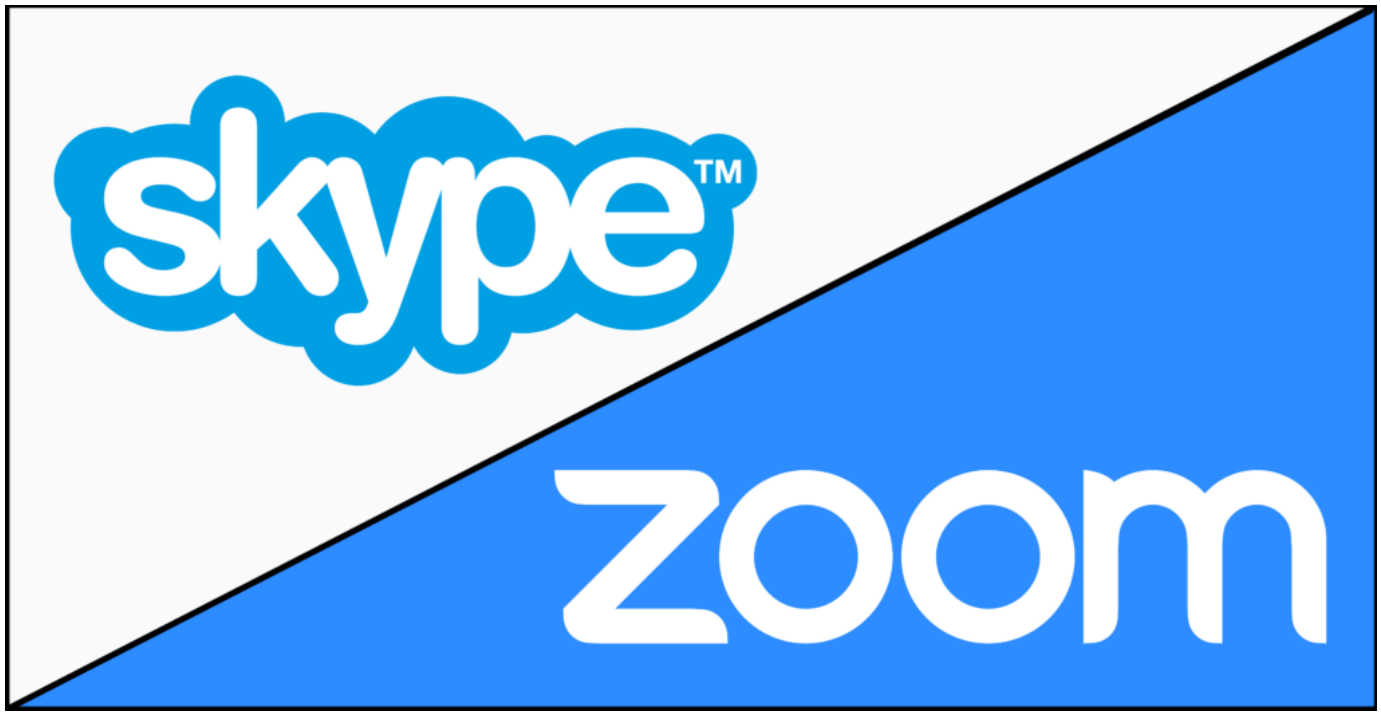
Simultaneous game & Sequential game have two different equilibria

First Mover Advantage Examples



amazon

Being the First is not always an Advantage



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Signaling 5 mts. https://www.youtube.com/watch?v=F8ZHZRMSxKg



Asymmetry of Information and Signaling Theory

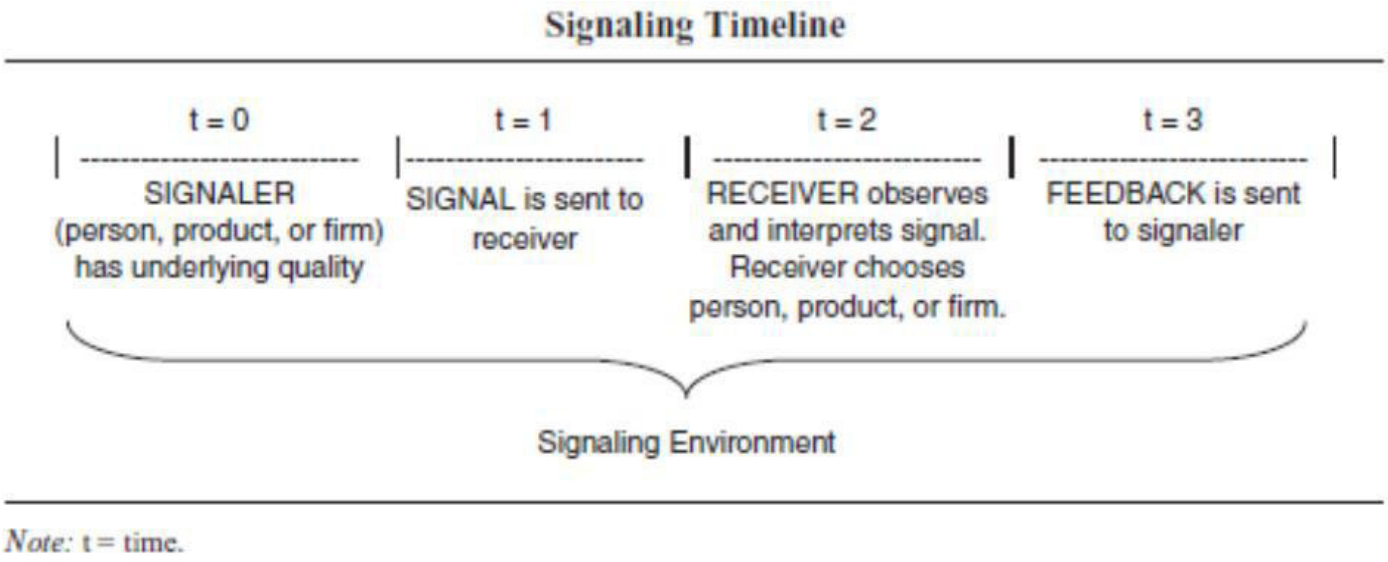
Signaling theory is useful for describing behavior when two parties (individuals or organizations) have access to different information.

- Typically, one party, the sender, must choose whether and how to **communicate (or signal) that information**, and the other party, the receiver, must choose how to interpret the signal.
- Signaling theory holds a prominent position in a variety of management literatures, including strategic management, entrepreneurship, and human resource management.

George Akerlof, Michael Spence, and Joseph Stiglitz received the 2001 Nobel Prize in Economics for their work in information economics.

From the Signaler to the Feedback

- The signal could create **commitment** and if the feedback is positive it also **creates trust**
  - **Collusive Strategies**
  - **Dynamic Models to sustain NE over time**



Note: t = time.

Entry Dynamics

Example: Boeing vs. Airbus

- Airbus announced its plan of entering the industry of 'very-large-aircrafts' by committing to invest 11.9 billion dollars for the launch of the A380
- Boeing tried to deter the entry of Airbus by threatening to produce a larger version of its 747
- Airbus did not believe in Boeing's threat and build a A380



- ***Would Boeing forecast the entry of Airbus?***
- ***Which alternative strategies did Boeing have to prevent the entry?***
- ***Which strategies did Boeing have to react to the entry?***

Thanks a lot!

Any questions?

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